

Project: Phylogenetic Analysis and Biological Data Visualization

Project Overview

In modern bioinformatics, scientists analyze DNA sequences to understand evolutionary relationships between organisms. These analyses are often combined with experimental measurements to provide a more complete biological interpretation.

In this project, you will build a complete data analysis pipeline using Python. You will integrate biological metadata (mass and volume) with DNA sequence data to construct a phylogenetic tree and generate scientific visualizations.

Learning Objectives

By completing this project, you will learn how to:

- Work with biological datasets in CSV and FASTA formats
- Clean and preprocess data using Python
- Compute derived scientific values (concentration in g/cm³)
- Compare DNA sequences using identity-based distance
- Construct a phylogenetic tree using the Neighbor Joining method
- Visualize biological data using different types of plots
- Interpret both experimental and evolutionary results

Project Data

CSV File (data.csv)

sample_id	species	mass	volume
S1	Escherichia coli	1.2	0.5
S2	Staphylococcus aureus	1.0	0.4
S3	Bacillus subtilis	1.3	0.6
S4	Aspergillus fumigatus	0.9	0.3
S5	Candida albicans	1.1	0.5

FASTA File (projectsequences.fasta)

```
>S1
ATGCTAGCTAGC
>S2
ATGCGAGCTAGC
>S3
ATGCTAGATAGC
>S4
ATGCGTGCTAGC
>S5
ATGCTAGCTTGC
```

Project Tasks

Task 1: Load and Prepare Data

- Load the CSV file using pandas
 - Clean column names
 - Display the dataset
-

Task 2: Compute Concentration

- Calculate concentration using the formula:

$\text{concentration} = \text{mass} / \text{volume}$

- Add a new column named `concentration`
 - Express the result in g/cm^3
-

Task 3: Load DNA Sequences

- Read sequences from the FASTA file using Biopython
 - Display sequence identifiers and sequences
-

Task 4: Create Alignment Object

- Build a Multiple Sequence Alignment object
- Prepare sequences for comparison

Task 5: Compute Distance Matrix

- Use identity-based distance with:

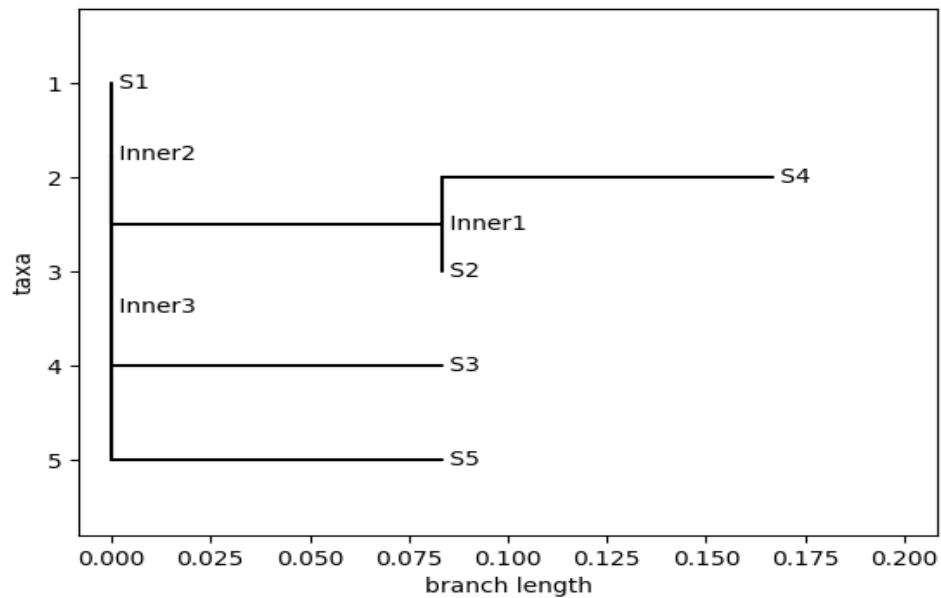
DistanceCalculator("identity")

- Understand that distance = 1 – similarity
-

Task 6: Construct Phylogenetic Tree

- Use the Neighbor Joining method
- Display the tree in ASCII format
- Visualize the tree graphically

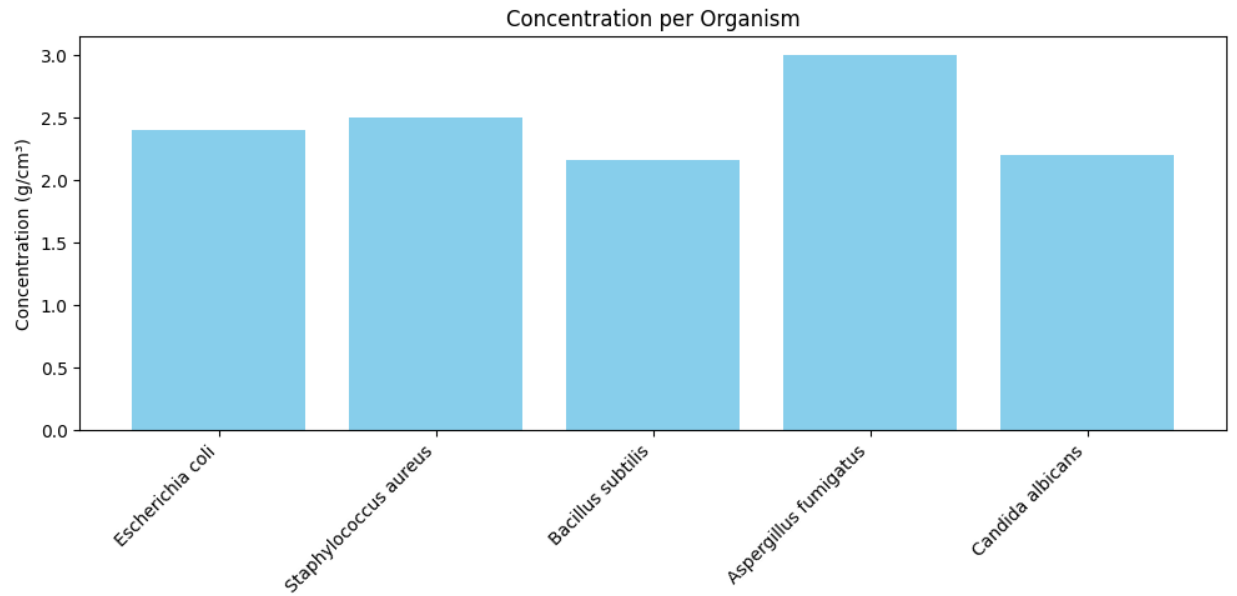
Phylogeny



Task 7: Visualize Data

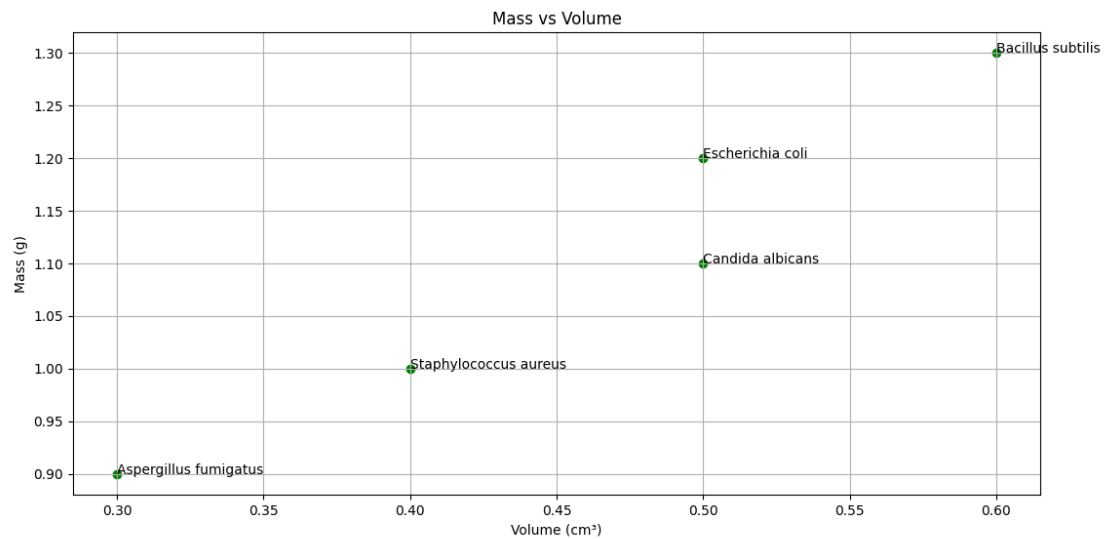
Bar Chart

- Compare concentration values between organisms



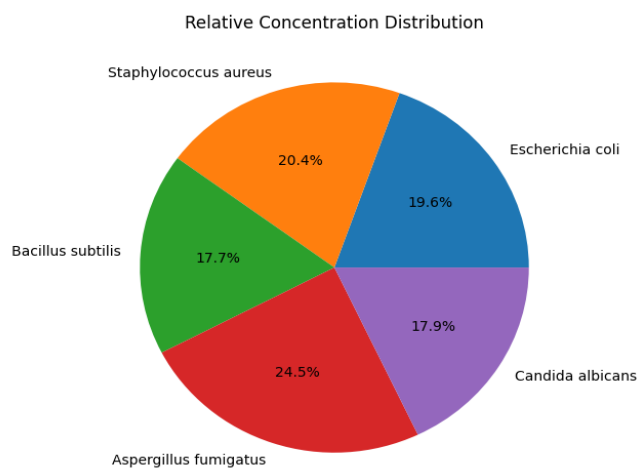
Scatter Plot

- Plot mass versus volume
- Analyze the relationship



Pie Chart

- Show the relative contribution of each organism



the line:

```
plt.pie(df["concentration"], labels=df["species"], autopct="%1.1f%%")
```

does:

1. Takes all concentration values:
 - [2.4, 2.5, 2.17, 3.0, 2.2]
2. Computes total:
 - $2.4 + 2.5 + 2.17 + 3.0 + 2.2 = 12.27$
3. Converts each value into percentage:

Example:

$$\frac{2.4}{12.27} \times 100 \approx 19.6\%$$

So each slice shows:

“How much this organism contributes to the total concentration pool”

Sample Outputs

Metadata Output

```
=== METADATA ===
  sample_id      species  mass  volume
0         S1  Escherichia coli   1.2    0.5
1         S2  Staphylococcus aureus  1.0    0.4
2         S3   Bacillus subtilis   1.3    0.6
3         S4  Aspergillus fumigatus  0.9    0.3
4         S5   Candida albicans   1.1    0.5
```

Concentration Output

```
=== CONCENTRATION (g/cm^3) ===
      species  concentration
0  Escherichia coli         2.40
1  Staphylococcus aureus         2.50
2   Bacillus subtilis         2.17
```

3	Aspergillus fumigatus	3.00
4	Candida albicans	2.20

Distance Matrix (Example)

```
=== DISTANCE MATRIX ===
S1  0.000000
S2  0.166667  0.000000
S3  0.083333  0.166667  0.000000
S4  0.166667  0.083333  0.166667  0.000000
S5  0.083333  0.166667  0.083333  0.166667  0.000000
```

Phylogenetic Tree (ASCII Example)

```

      |_____ Escherichia coli
      |
      |_____ Bacillus subtilis
      |
      |_____ Candida albicans
      |
      |_____ Staphylococcus aureus
      |
      |_____ Aspergillus fumigatus
```

(The exact structure may vary slightly.)

Graph Outputs

- Bar chart showing concentration per organism
 - Scatter plot showing mass versus volume
 - Pie chart showing proportional distribution
-

Analysis Questions

1. Which organisms are most genetically similar?
2. Which organism has the highest concentration?
3. Do genetically similar organisms share similar physical properties?
4. What does the phylogenetic tree suggest about relationships between bacteria and fungi?
5. Is there a relationship between mass and volume?

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from Bio import SeqIO
from Bio.Align import MultipleSeqAlignment
from Bio.Phylo.TreeConstruction import DistanceCalculator, DistanceTreeConstructor
from Bio import Phylo

# =====
# 1. LOAD METADATA (CSV)
# =====

df = pd.read_csv("data.csv")

# Clean ALL column names
df.columns = df.columns.str.strip().str.lower()

print("\n=== METADATA ===")
print(df)

# =====
# 2. COMPUTE CONCENTRATION (g/cm^3)
# =====

df["concentration"] = df["mass"] / df["volume"]
```



```

print("\n=== CONCENTRATION (g/cm^3) ===")
print(df[["species", "concentration"]])

# =====

# 3. LOAD FASTA SEQUENCES
# =====

records = list(SeqIO.parse("projectsequences.fasta", "fasta"))

print("\n=== RAW SEQUENCES ===")
for r in records:
    print(r.id, r.seq)

# =====

# 4. MULTIPLE SEQUENCE ALIGNMENT
# =====

alignment = MultipleSeqAlignment(records)

# =====

#5. DISTANCE MATRIX (IDENTITY)
# =====

calculator = DistanceCalculator("identity")
dm = calculator.get_distance(alignment)

print("\n=== DISTANCE MATRIX ===")

```

```

print(dm)

# =====

# 6. PHYLOGENETIC TREE (NJ)

# =====

constructor = DistanceTreeConstructor()
tree = constructor.nj(dm)

print("\n=== PHYLOGENETIC TREE (ASCII) ===")
Phylo.draw_ascii(tree)

# =====

# 7. GRAPHICAL TREE

# =====

Phylo.draw(tree)

# =====

# 8. BAR CHART (CONCENTRATION)

# =====

plt.figure(figsize=(10, 5))
plt.bar(df["species"], df["concentration"], color="skyblue")

plt.xticks(rotation=45, ha="right")
plt.ylabel("Concentration (g/cm³)")
plt.title("Concentration per Organism")

```

```
plt.tight_layout()
```

```
plt.show()
```

```
# =====
```

```
# 9. SCATTER PLOT (MASS vs VOLUME)
```

```
# =====
```

```
plt.figure(figsize=(6, 5))
```

```
plt.scatter(df["volume"], df["mass"], color="green")
```

```
for i, species in enumerate(df["species"]):
```

```
    plt.text(df["volume"][i], df["mass"][i], species)
```

```
plt.xlabel("Volume (cm3)")
```

```
plt.ylabel("Mass (g)")
```

```
plt.title("Mass vs Volume")
```

```
plt.grid(True)
```

```
plt.show()
```

```
# =====
```

```
# 10. PIE CHART (CONCENTRATION DISTRIBUTION)
```

```
# =====
```

```
plt.figure(figsize=(6, 6))
```

```
plt.pie(df["concentration"], labels=df["species"], autopct="%1.1f%%")
```

```
plt.title("Relative Concentration Distribution")
```

```
plt.show()
```